

MorphDet: Towards the Detection of Morphing Attacks

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Abstract—Biometric authentication systems have become an inevitable part of the society. They are based on the primary traits of an individual that are unique and hard to forge or manipulate by simple means. However, the unprecedented growth of technology has enabled the access of so many advanced tools that could be used for forging these traits. In this paper, we focus on the face morphing attacks. A basic pipeline is used to generate morphed attacks. A face morph detection model based on Resnet-152 is proposed and validated through exhaustive experiments. A dataset of 28,890 images is also contributed to conduct the experiments for varied scenarios, including simple face images, faces with beards, faces with eyeglasses, and a combination of beard and eyeglasses. Comparative performance analysis is done with the other state-of-the-art models i.e. Alexnet and VGG-16 and the proposed framework is found to outperform them.

Index Terms—Face morphing, Morphing attack detection, Biometrics.

I. INTRODUCTION

Biometric traits are frequently used to identify and authenticate a person using a set of recognizable and verifiable data unique and specific to an individual. The most common uses of biometrics in everyday life include airport security, law enforcement, mobile access and authentication, banking, schools, border, travel, migration control and many more [4]. Among these applications, some could be very crucial, for instance, allowing a person across country borders where the facial characteristics are compared with a reference database to verify a person's identity and grant/deny the access [7], [3].

Although Facial Recognition Systems (FRS) have been quite effective in distinguishing an individual from other subjects but the unprecedented growth of technology is now making this identification quite challenging. Many face morphing attacks has been reported recently to fool these FRS. On "Tucker Carlson Tonight," a digitally manipulated image was broadcast by Fox News. Epstein's upper body was altered with Reinhart's, which quickly gained popularity on Twitter [8]. Other example is an old woman have been harassed by online loan apps. She received her morphed images on WhatsApp from an unknown person [1]. Face morphing can be simply defined as the blending of facial traits from two similar looking faces to result in a morphed face that resembles both the faces as shown in Fig. 1. This morphed image can be exploited to bypass or fool the FRS. In today's contemporary world, 'Morphing' is almost everywhere, be it the film industry, car-

toons and animations industry, social media platforms floating fake news, etc. Even a naive person can successful frame a morph image without putting much efforts owing to the easy availability of free tutorials/software.

Various techniques for the detection of the morphed faces have been proposed in the literature. These techniques can be broadly divided into three main categories: (a) texture based, (b) image quality based, and (c) deep learning based approaches. Algorithms-based on texture features such as Binary Statistical Image Features (BSIF), Local Phase Quantization (LPQ) [13] and Local Binary Pattern (LBP) [13], [19] and it's variants are introduced to detect the digital version of morphed faces. Among these texture-based techniques, the use of LBP and BSIF has indicated consistent morphed face detection performance. The image quality-based methods are designed to quantify the variation in the compression artifacts and morphing noise introduced during the process of morph generation [11], [5]. Deep learning approaches, especially based on the pre-trained CNN architecture like AlexNet, VGG, ResNet, GoogleNet, InceptionV3, etc. have shown improvement over texture-based approaches [14] [17]. However, these neural networks face the issue of overfitting, problem of vanishing gradient, increase of depth in the layers make the prediction of the earlier layers very insignificant, more efforts required to optimize the huge parameter space that incurs the chances of higher training error, etc.

In this paper, a framework is proposed to detect whether an image is authentic or morphed. The Convolutional Neural Network (CNN) based classifier is exploited to perform this discrimination. Resnet-152 model is employed towards achieving this objective. A dataset of 28,890 images has been contributed to conduct the experiments for varied scenarios: The model detects whether the image is original or morphed in the first scenario, resulting in a binary classification. The second scenario goes a step further, determining if the image is original or altered, as well as the individual class the suspected image belongs to. As a result, the second scenario covers three classes: Individual 1, Individual 2, and the morphed. The third scenario expands this to four subjects and six different morphed image combinations from combining any two individuals at the same time. In total, ten classifications are taken into account here. Comparative performance analysis is done with the other pre-trained and popular models trained



(a) Image 1



(b) Morph Image



(c) Image 2

Fig. 1: Morphed image of two input images

from the scratch. The major contributions of the paper can be summarized as follows:

- **Dataset:** We created a dataset of 28,460 images to conduct experiments covering various scenarios for detection of face morphing. Face images of four different subjects: Aamir Khan's image, Amitabh Bachchan's image, Prabhas' image and Salman Khan's image were collected and cropped along the outer boundary of the standard 68 landmark points as shown in Fig. 3a III-A. Thereafter, morphing was done using these cropped face images using the generalized face morphing algorithm III.
- **Experiment scenarios:** The performance of the proposed framework has been validated with an exhaustive set of experiment scenarios. In the first scenario, the model detects whether the image is an original image or a morphed image, hence a binary classification. Second scenario goes one level deeper and can distinguish whether the image is original or morphed along with detecting which individual class the suspected image belongs to. Hence, second scenario considers three classes: Individual 1 as first class, Individual 2 as second class, and the morphed as the third class. To perform an exhaustive testing, we further considered varied sub-scenarios here: first considering only simple face images, second containing bearded face images, third containing face images wearing eyeglasses and fourth containing faces with beard as well as eyeglasses. Third scenario extends this for four subjects and six possible combinations of morphed images from combining any two individual at a time. In total, 10 classes are considered here.
- **Comparative performance analysis:** The performance of the proposed model is compared with other pre-existing models: Alexnet and VGG16.

The rest of the paper is organized as follows: Section II describes the Related work. Section III presents the Morphing

pipeline. Section IV presents different detection scenarios, data set for the respective scenarios and experiments conducted. Section IV-D presents the comparative performance analysis of our model with other pre-existing state-of-the-art approaches. Section V brings the paper to a conclusion.

TABLE I: Variable list

Variable	Definition
I_1	Image 1
I_2	Image 2
S_{I_1}	Set of Landmark points in image 1
S_{I_2}	Set of Landmark points in image 2
I_M	Morph Image
S_{I_M}	Set of Landmark points in Morph Image
T_1	Triangle in image 1
T_2	Triangle in image 2
T_M	Triangle in Morph image

II. RELATED WORK

In this section, we briefly discuss the relevant literature related to face morphing:

Jager et al. described one of the first studies on human perception analysis of morphed images, in which numerous experiments were conducted to evaluate human observers' capacity to detect face morphing and its dependence on various parameters i.e., different alpha/morphing factors [9]. Furthermore, this research relied just on a single image and provided no reference images for human observers. Kramer et al. provided a similar analysis, where a single image was provided before requesting a decision on morphing [10]. Despite the fact that their bench-marking mechanisms were different, both studies observed difficulty in spotting morphing face image for human observers.

Ulrich et al. presented a conceptual categorization and metrics for an evaluation of face morphing and detection

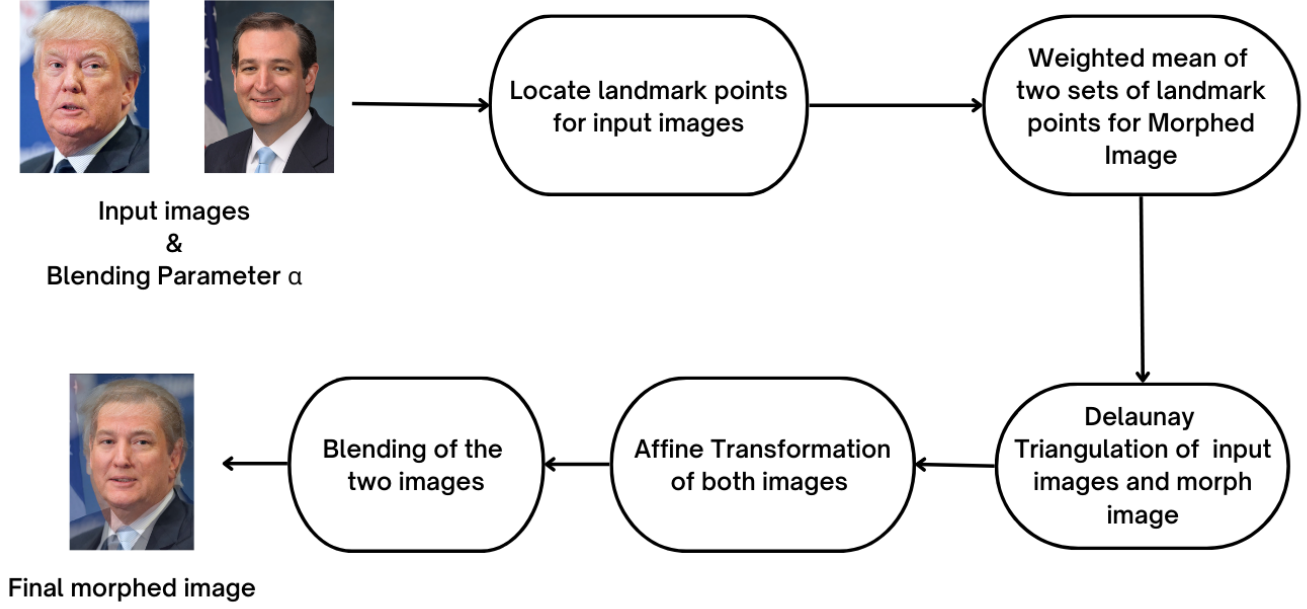


Fig. 2: Flowchart of face morphing

methods [15]. In addition, they pointed out most relevant issues and challenges in the field. One of them is the quality of the morphed images in order to enable statistically significant testing of developed morphing attack detection methods under realistic conditions. Seibold et al. proposed reflection analysis method for detection of face morphing [16]. It was based on reflection of lights in eyes and synthesize the highlights on the nose using 3-D model. Based on the synthesis highlights, they were able to distinguish whether the image is of individual or morphed. It lacked training and classification and required manual intervention to classify the image.

Luuk et al. proposed face morphing databases with varying characteristics [19]. Experiments with mixed datasets suggested that morphing detection can be made more robust if trained on multiple datasets. They said morphing detection methods should be tested extensively on multiple datasets obtained from different sources and morphing methods and a range of image manipulations.

Noting the limitations of human observers, a number of automatic morphing attack detection (MAD) approaches have been proposed in the recent past. The available MAD techniques can be classified into two major types: (a) single image-based MAD (S-MAD) and (b) differential image-based MAD (D-MAD). The goal of S-MAD is to effectively detect a face morphing attack based on a single image presented to the algorithm. The objective of D-MAD approaches is to make a decision regarding whether a suspect image is morphed or bonafide when a corresponding image captured in a trusted environment is available. Single image MAD procedures can depend heavily on the training data used and

can only detect the artefacts learned during training. This can greatly limit the generalizability of these methods. For these reasons, differential MAD procedures are generally to be seen as more promising [6].

Tom Neubert introduced novel detection approach for face morphing detection based on image degradation [12]. The accuracy for this model came out to be 91.3 % in laboratory conditions. After that he built six different datasets to test conditions. The accuracy for this model was 85.9% but for real world scenario, the accuracy was 68.4%, which is not very promising for real-world applications. In this paper, we are further extending the face morph detection work using CNN and RESNET-152 model. Here, we use a single-image based MAD considering different scenarios to detect the possible artifacts.

III. MORPHING PIPELINE

This section discusses the generalized face morphing algorithm step by step. The overview of the basic steps is presented briefly in the flowchart in Fig. 2.

A. Locate landmark points

The first basic step of face morphing is to locate landmark points for both the images that are intended to morph. Landmark points are the primary features of a face such as eyes, nose, lips, eyebrows, etc. These are used to identify where the human face is located in the whole image and track key-points from a human face. Consider two images shown in Fig. 1 (a) and Fig. 1 (c) as I_1 and I_2 . All the variables used are listed in Table I.

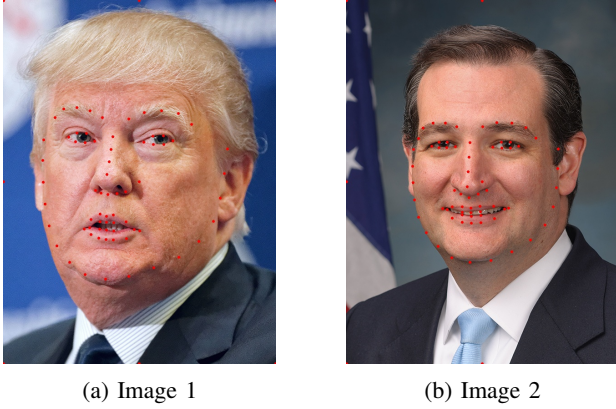


Fig. 3: Landmark Points of two Images

There are 68 landmark points in the image which covers the region around eyes, eyebrows, nose, mouth, chin and jaw. Also, since this is a one-to-one correspondence, the number of landmark points in I_1 will be equal to the number of landmark points in I_2 . So, we have two sets S_{I_1} and S_{I_2} of landmark points for images I_1 and I_2 respectively.

$$S_{I_1} = p_1, p_2, p_3, \dots, p_k \quad (1)$$

$$S_{I_2} = q_1, q_2, q_3, \dots, q_k \quad (2)$$

Further, calculate a weighted mean of the points in the two sets, based on the value of α and obtain another set of landmark points for the morph image I_M . Let this set of points be denoted by S_{I_M} .

So, $S_{I_M} = i_1, i_2, i_3, \dots, i_k$
where

$$i_k = (1 - \alpha) \cdot p_k + \alpha \cdot q_k \quad (3)$$

The landmark points for I_1 and I_2 are shown in Fig. 3.

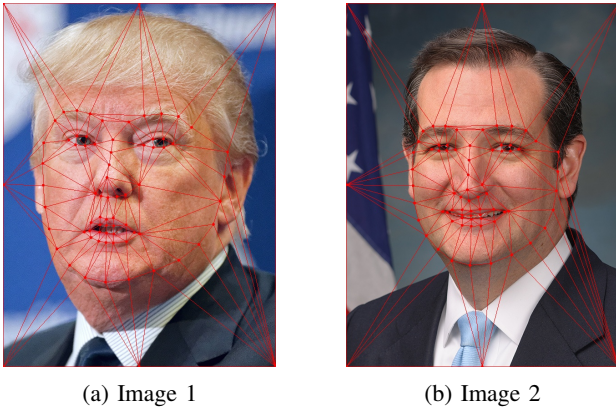


Fig. 4: Delaunay triangulation of two images

B. Delaunay triangulation

Now, find delaunay triangles using a set of landmark points S_M calculated in the above step which divides the input images and the morph image into many small triangles.

Further, the triangulation is performed for the other two sets of point S_{I_1} and S_{I_2} , giving a one to one correspondence between triangles from image I_M . In Fig. 4, we can see the delaunay triangulation of both the images.

C. Affine transformation

Here, the transformation of the triangles of image I_1 and I_2 to the dimensions of corresponding triangles in morph image I_M is done.

For this, select a triangle T_1 from image I_1 , its corresponding triangle T_M in the morph image I_M , and calculate the affine transform that maps the three corners of the triangle T_1 in image I_1 to the three corners of the corresponding triangle T_M in the morphed image I_M . Then, apply affine transform to all the triangles which applies the transformation matrix pixel by pixel in image I_1 to get the warped image I_1' . Similarly, get the warped image I_2' using image I_2 .

D. Blending

After the alignment of the two contributing images, blend the two images to get the morph image and the most frequent way of blending for face morph creation is alpha blending.

The equation for alpha blending is given as:

$$M(x, y) = (1 - \alpha) \cdot I_1'(x, y) + \alpha \cdot I_2'(x, y) \quad (4)$$

where I_1' is warp image of image I_1 and I_2' is warp image of image I_2 .

The morphed image of the two contributing images is shown in Fig. 1.

IV. DETECTION OF FACE MORPHING ATTACKS

In this section, details of the various experiment scenarios, description of the dataset collected, and the findings pertaining to the experiment scenarios are discussed. In order to identify face morphing attacks, the CNN-based RESNET-152 model is exploited and trained using batch size of 64, number of epochs as 30, and learning rate as 0.001.

A. Experiment scenarios

This sub-section describes the experiment scenarios:

- 1) **First Scenario:** The first scenario detects whether a given image is original or morphed. It is a binary classification and considers two classes: original image and morph image.

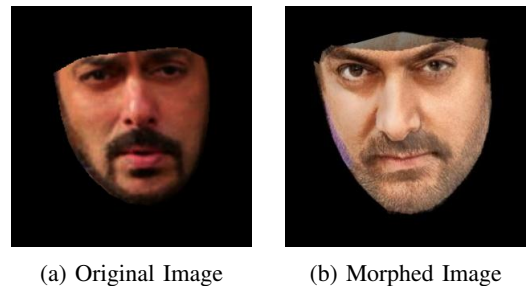
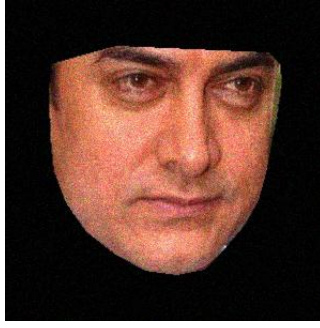


Fig. 5: First scenario



(a) Amitabh's Image



(b) Aamir's Image



(c) Morphed Image

Fig. 6: Second scenario sub-dataset 1



(a) Aamir's Image



(b) Salman's Image



(c) Morphed Image

Fig. 7: Second scenario sub-dataset 2



(a) Salman's Image



(b) Amitabh's Image



(c) Morphed Image

Fig. 8: Second scenario sub-dataset 3

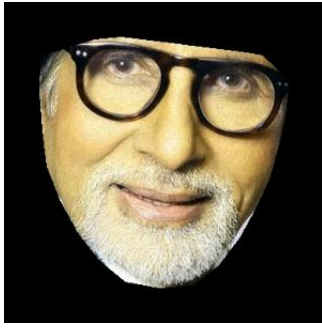
- 2) **Second Scenario:** The second scenario goes a step further by identifying the class to which the suspected image belongs as well as detecting if the image is original or altered. As a result, the second scenario covers three classes: individual 1 as the first class, individual 2 as the second class, and the morphed as the third class. Here, we analyzed four sub-scenarios in order to carry out thorough testing: the first had only plain face photographs, the second contained bearded face images, the third contained eyeglass-wearing face images, and the fourth contained faces with beards and eyeglasses.

- 3) **Third scenario:** The third scenario extends second sce-

nario to four subjects and six different morphed image combinations by combining any two individuals at a time. In total, ten classes are considered here.

B. Database

This sub-section describes the dataset collected for carrying out the experiment for the three aforementioned scenarios. In the dataset, every image has a frontal view of a person's face, as necessary for passport photos. The captured individual has open eyes, a closed or open mouth, eyeglasses, a beard face, varied perspectives, and different brightness and contrast settings.



(a) Amitabh's Image

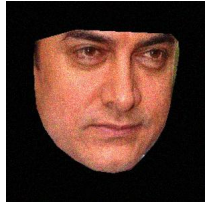


(b) Prabhas' Image

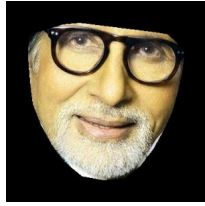


(c) Morphed Image

Fig. 9: Second scenario sub-dataset 4



(a) Aamir's Image



(b) Amitabh's Image



(c) Prabhas' Image



(d) Salman's Image



(e) Morphed Image



(f) Morphed Image



(g) Morphed Image



(h) Morphed Image



(i) Morphed Image



(j) Morphed Image

Fig. 10: Third scenario

- 1) **First Scenario:** For this scenario, we used a total of 18488 images for training, 4623 images for validation, and 5779 photos for testing. This scenario involves identifying if the image is a morph or the original.
- 2) **Second Scenario:** Here, we have a dataset of 28,890 total images. Further to conduct thorough testing, we examined four sub-scenarios: simple face images, face with only beard images, face with only eyeglasses images and face with both beard as well as eyeglasses images. For simple face images sub-dataset, total of 3834 training images, 959 validation images and 1198 testing images were used. For face with only beard images sub-dataset, total of 5558 training images, 1389 validation images and 1736 testing images were used. For face with only eyeglasses images subdataset, total of 1682 training images, 336 validation images and 421 testing images were used. For face with beard and eyeglasses images subdataset, total of 7268 training images, 1817 validation images and 2271 testing images were used.
- 3) **Third Scenario:** In this scenario, total of 18488 training Images, 4623 validation images and 5779 testing Images were used. In total, ten classifications are taken into

account here.

C. Experiment results

This sub-section presents the results of the experiments conducted to validate the proposed detection model. The accuracy of the classification scenarios are summarized in Table III.

- 1) **First scenario:** Here, total number of testing samples considered were 5779, out of which 5221 testing samples were classified correctly. Hence, the accuracy obtained here is 90.37%.
- 2) **Second scenario:** Here, we examined four sub-scenarios: the first with only plain face images, the second with bearded face images, the third with face images wearing eyeglasses, and the fourth with beards and eyeglasses. The result of those 4 sub-scenarios is as follows.
 - a) For sub-scenario 1, we used simple face images without eyeglasses and beard. For this, we took 4 different individuals images and their morph combinations. At first, we conducted experiment by taking any two individual's image and their morph and determined the accuracy. For instance,

TABLE II: Second scenario accuracy table

Sub-scenarios	Individual 1	Individual 2	Morphed	Resnet-152	VGG16 (Pre-trained) [18]	Alexnet (Scratch) [2]
Simple Face Images	Aamir	Amitabh	Morph	83.68%	93.33%	61.05%
	Aamir	Prabhas	Morph	88.73%	77.46%	53.52%
	Aamir	Salman	Morph	76.49%	73.67%	76.61%
	Amitabh	Salman	Morph	84.40%	84.62%	73.18%
	Amitabh	Prabhas	Morph	93.47%	86.96%	77.53%
	Prabhas	Salman	Morph	96.86%	92.95%	89.03%
Beard Face Images	Aamir	Amitabh	Morph	85.01%	76.57%	52.58%
	Aamir	Prabhas	Morph	87.87%	87.65%	46.49%
	Aamir	Salman	Morph	83.23%	92.95%	65.52%
	Amitabh	Salman	Morph	85.65%	81.17%	58.30%
	Amitabh	Prabhas	Morph	94.12%	88.08%	67.88%
	Prabhas	Salman	Morph	88.96%	80.16%	61.60%
Eyeglasses Face Images	Aamir	Amitabh	Morph	97.29%	76.38%	83.78%
	Aamir	Prabhas	Morph	94.36%	84.51%	53.52%
	Aamir	Salman	Morph	97.77%	93.21%	66.66%
	Amitabh	Salman	Morph	95.36%	69.54%	53.64%
	Amitabh	Prabhas	Morph	97.61%	79.69%	59.52%
	Prabhas	Salman	Morph	98.36%	81.97%	66.39%
Beard and Eyeglasses Face Images	Aamir	Amitabh	Morph	95.52%	89.31%	56.21%
	Aamir	Prabhas	Morph	88.02%	91.55%	53.52%
	Aamir	Salman	Morph	90.18%	96.93%	56.44%
	Amitabh	Salman	Morph	96.31%	92.03%	55.41%
	Amitabh	Prabhas	Morph	93.66%	94.07%	56.58%
	Prabhas	Salman	Morph	95.63%	86.27%	68.17%

TABLE III: Comparative analysis of the proposed model with other state-of-the-art approaches

	Resnet-152	VGG16 [18] (Pre-trained)	Alexnet [2] (From scratch)
First Scenario	90.37%	84.38 %	93.21%
Second Scenario sub-dataset 1	87.27%	84.83%	71.83%
Second Scenario sub-dataset 2	87.47%	84.43%	58.72%
Second Scenario sub-dataset 3	96.79%	80.88%	63.91%
Second Scenario sub-dataset 4	93.22%	91.69%	57.72%
Third Scenario	45.53%	45.12%	50.72 %

we took Aamir's and Amitabh's image and their morph and determined the accuracy. Then we took 2 other individual's combination, say Aamir and Prabhas, and their morph and find the accuracy. Similarly, we conducted in total 6 experiments as described in Table II. Lastly, we calculated the average accuracy of the 6 experiments which turned out to be 80.55%.

- b) For sub-scenario 2, we used beard face images without eyeglasses. For this also, we followed the same approach as for the sub-scenario 1 and averaged the accuracy over all the iterations and got an accuracy of 86.01%.
- c) For sub-scenario 3, we used eyeglasses face images without beard. Similar iterative process done here also as for earlier sub-scenarios and calculated the average accuracy to be 74.22%.
- d) For sub-scenario 4, we used face images having both beard and eyeglasses. Similar iterative process done here also as for earlier sub-scenarios and calculated the average accuracy to be 86.78%.

- 3) **Third Scenario:** In this scenario, there are 10 classes for classification. Four classes for individuals and six classes for the morphed images. Here, as shown in Fig. 10, class 0 is Aamir khan (Fig. 10a), class 1 is Amitabh bachchan (Fig. 10b), class 2 is Prabhas (Fig. 10c), class 3 is Salman khan (Fig. 10d), class 4 is morph image of Aamir khan and Amitabh bachchan (Fig. 10e), class 5 is morph image of Aamir khan and Prabhas (Fig. 10f), class 6 is morph image of Aamir khan and Salman khan (Fig. 10g), class 7 is morph image of Amitabh bachchan and Prabhas (Fig. 10h), class 8 is morph image of Amitabh bachchan and Salman khan (Fig. 10i), and class 9 is morph image of Prabhas and Salman khan (Fig. 10j).

The total number of testing samples used were 5779, out of which, the number of correctly classified were 2631. Hence, the accuracy came out to be 45.53%.

D. Comparative performance analysis

To validate the performance of the proposed Resnet-152 model, it is compared with pre-trained VGG16 and Alexnet CNN models. Pre-existing models VGG16 and Alexnet were trained and tested with our dataset. For Resnet-152, input image size was 256×256 . For the VGG16 model, the input image size was 224×224 , whereas for the Alexnet model, it was 227×227 . All the models were trained for 30 epochs, 64 batches, and a learning rate of 0.001. VGG16 is a 16-layer deep model implemented using transfer learning, whereas Alexnet is an eight-layer CNN model. There is no automatic way to select the best model for each dataset and domain. Using pre-train model gives us better results. The reason behind implementing alexnet from scratch was to reduce overfitting. We observed that Alexnet works well for binary classification, but as the number of classes are increased, the accuracy of the Alexnet model keeps on reducing. But on the whole, we can conclude that Resnet-152 model is relatively more efficient than the other models.

V. CONCLUSION

In this paper, a model is proposed to detect face morphing attacks. Various experiment scenarios, i.e. simple face morphs, face morphs with beards, face morphs with eyeglasses, and face morphs with a combination of beards and eyeglasses are considered to validate the proposed model. A dataset covering these scenarios is also contributed to carry out the experiments. Further, a comparative performance analysis using the dataset is done with the popular pre-existing CNN models: Alexnet and VGG16. As a whole, the proposed Resnet-152 has shown better performance in terms of accuracy. In future, we plan to extend this model for more possible attack scenarios and test the scalability with other available benchmark datasets.

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